

# Tianhao Hu

## RESEARCH INTEREST

**Numerical PDE,  
Machine Learning,  
Inverse Problems,  
Scientific Computing**

## CONTACT

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Building, The Chinese University of Hong Kong,  
Shatin, New Territories, Hong Kong**

## PERSONAL WEBSITE

Personal website:

<https://hhjc-web.github.io/>

Google Scholar:

<https://scholar.google.com/citations?user=D7NWc1QAAAAJ&>

GitHub: <https://github.com/hhjc-web>

## PUBLICATION LIST

- T. Hu, B. Jin, and Z. Zhou, Solving elliptic problems with singular sources using singularity splitting deep Ritz method, SIAM Journal on Scientific Computing, 45 (2023), pp. A2043-A2074.
- T. Hu, B. Jin, and Z. Zhou, Solving Poisson problems in polygonal domains with singularity enriched physics informed neural networks, SIAM Journal on Scientific Computing, 46 (2024), pp. C369-C398.
- T. Hu, B. Jin, and F. Wang, An iterative deep Ritz method for monotone elliptic problems, Journal of Computational Physics, 527 (2025), pp.113791.
- T. Hu, B. Jin, and Z. Zhou, Point source identification using singularity enriched neural networks, IMA Journal of Numerical Analysis, page draf129, 02 2026.
- T. Hu, X. Huang, B. Jin, Q. Quan, and Z. Zhou, Conditional Stability and Numerical Reconstruction of a Parabolic Inverse Source Problem Using Carleman Estimates, Inverse Problems, minor revision, 2026.
- T. Hu, B. Jin, and Y. Wang, Stability of Electrical Impedance Tomography with Anisotropies and its Application to the Deep Caldeón Method, SIAM Journal on Imaging Sciences, in press, 2026.
- T. Hu, G. Li, F. Wang, Y. Xu, and Z. Zhou. Dual variational neural network for the p-Laplace problem, arxiv.org: 2605.28291, submitted to ESAIM: Mathematical Modelling and Numerical Analysis, 2026.
- T. Hu, B. Jin, and Y. Wang. A Convolutional Deep Calderón Method for Electrical Impedance Tomography, submitted to Proceedings of The Royal Society, Series A, 2026.
- T. Hu, B. Jin, F. Wang, and Y. Xu. A Hybrid Iterative Deep Ritz Method for Elliptic Interface Problems, arxiv.org: 2609.27325, submitted to Computers and Mathematics with Applications, 2026.

## ONGOING PROJECTS

- T. Hu, B. Jin, and F. Wang, A Level-Set Random Feature Petrov–Galerkin Method for Elliptic and Parabolic Interface Problems.
- T. Hu and Z. Zhou, A recurrent Fourier Neural Network to Recovery Two Function-valued Coefficients in the Helmholtz Equation for Inverse Scattering Problems.
- T. Hu, B. Jin, Y. Wang, Learning Solution maps from Linear Wave equations to Nonlinear wave equations.

## EDUCATION

- 09/2019-06/2023, B.Sc., School of Mathematics, Jilin University.
- 08/2023-07/2027(estimated), Ph.D., Department of Mathematics, The Chinese University of Hong Kong.

## CONFERENCES

- Invited Speaker, talk "Solving Elliptic Problems with Singular Solutions Using Splitting Technique", at The Second Hong Kong SIAM Biennial conference, Hong Kong, 28<sup>th</sup> Aug – 1<sup>st</sup> Sep, 2023.
- Invited Speaker, talk "Identifying Point Sources in Elliptic Problems using Deep Neural Networks", at SIAM Conference on Computational Science and Engineering, Fort Worth, Texas, U.S., 3<sup>rd</sup>-7<sup>th</sup> Mar, 2025.
- Invited Speaker, talk "Point Source Identification using Singularity Enriched Neural Networks", at The 14<sup>th</sup> Annual Conference on Inverse Problems, Imaging, and Applications, Tianjin, China, 18<sup>th</sup>-21<sup>st</sup> Apr, 2025.
- Invited Speaker, talk "Dual variational neural network for the p-Laplace problem", at The 15<sup>th</sup> AIMS Conference, Athens, Greece, 6<sup>th</sup>-10<sup>th</sup> Jul, 2026.

## GRANTS

- 2026, National Natural Science Foundation of China, Project 125B2022, ¥ 300,000, PI.

## RESEARCH VISITS

- 10/2026-12/2026, Center for Computational Mathematics, Flatiron Institute, New York, Host: Prof. Jiequn Han.

## AWARDS

### SCHOLARSHIPS

- |                    |                                          |                                        |
|--------------------|------------------------------------------|----------------------------------------|
| (i) 2020,2021,2022 | National scholarship                     | Ministry of Education, China           |
| (ii) 2022          | Samsung Scholarship                      | Samsung (China) Investment Corporation |
| (iii)2022          | Hongqi Scholarship                       | China FAW Group Corporation            |
| (iv)2023           | CUHK Vice-Chancellor's Ph.D. Scholarship | The Chinese University of Hong Kong    |
| (v) 2024           | Madam Wong Siew Khoon Scholarship        | The Chinese University of Hong Kong    |

### RECOGNITIONS

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|----------|-------------------------------------------------------------|----------------------------------------------|
| (i) 2026 | CAST Youth Talent Cultivation Project – PhD Special Program | China Association for Science and Technology |
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## TEACHING EXPERIENCE

- Teaching Assistant, MATH 1510G Calculus for Engineers, 2023 Fall.
- Teaching Assistant, MATH 3270B Ordinary Differential Equations, 2024 Summer.
- Teaching Assistant, MATH 3230A Numerical Analysis, 2024 Fall.
- Teaching Assistant, MATH 3340 Mathematics of Machine Learning, 2025 Spring.
- Teaching Assistant, MMAT 5391 Mathematical Theories of Machine Learning, 2025 Fall.
- Teaching Assistant, MATH 3340 Mathematics of Machine Learning, 2026 Spring.
- Teaching Assistant, MMAT 5392 Mathematical Principles of Artificial Intelligence, 2027 Spring.
- Teaching Assistant, MATH 3340 Mathematics of Machine Learning, 2027 Spring.